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Joint lab on Safety and Security of AI

Computer Vision Technologies and Biometrics

Multimodal biometrics

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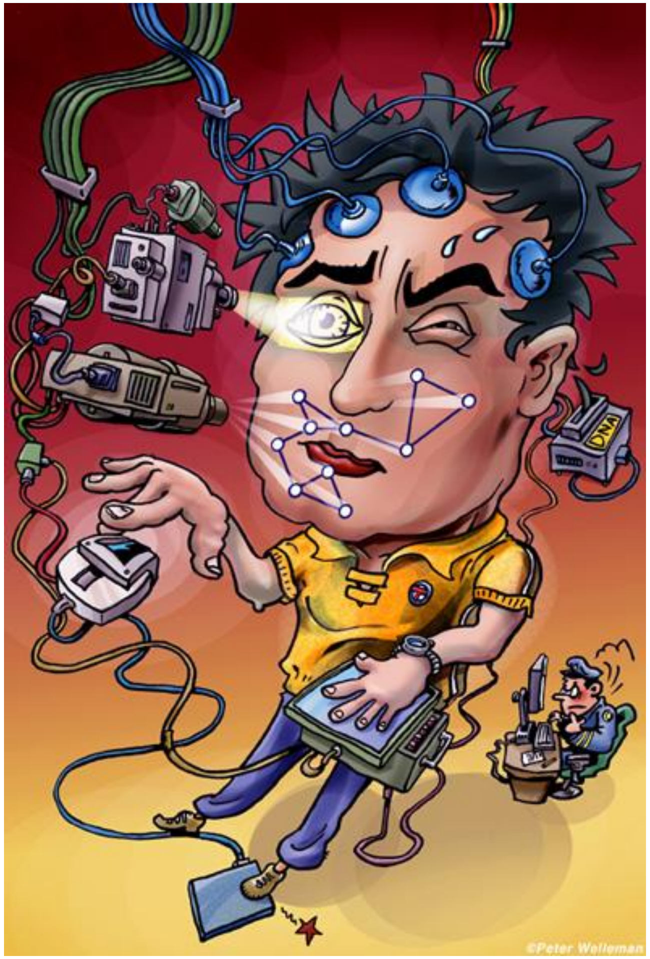
Outline of the talk

- **Multi-modal biometric systems**
  - properties
  - pros and cons
- **Fusion approaches**
  - Feature-level fusion
  - Measurement-level fusion
  - Decision-level modern challenge
- **Fusion architectures**
  - Parallel
  - Serial

Unformal meaning

- **In general, multi-modal biometrics is referred to methods employing the joint use of multiple inputs deriving from biometrics**



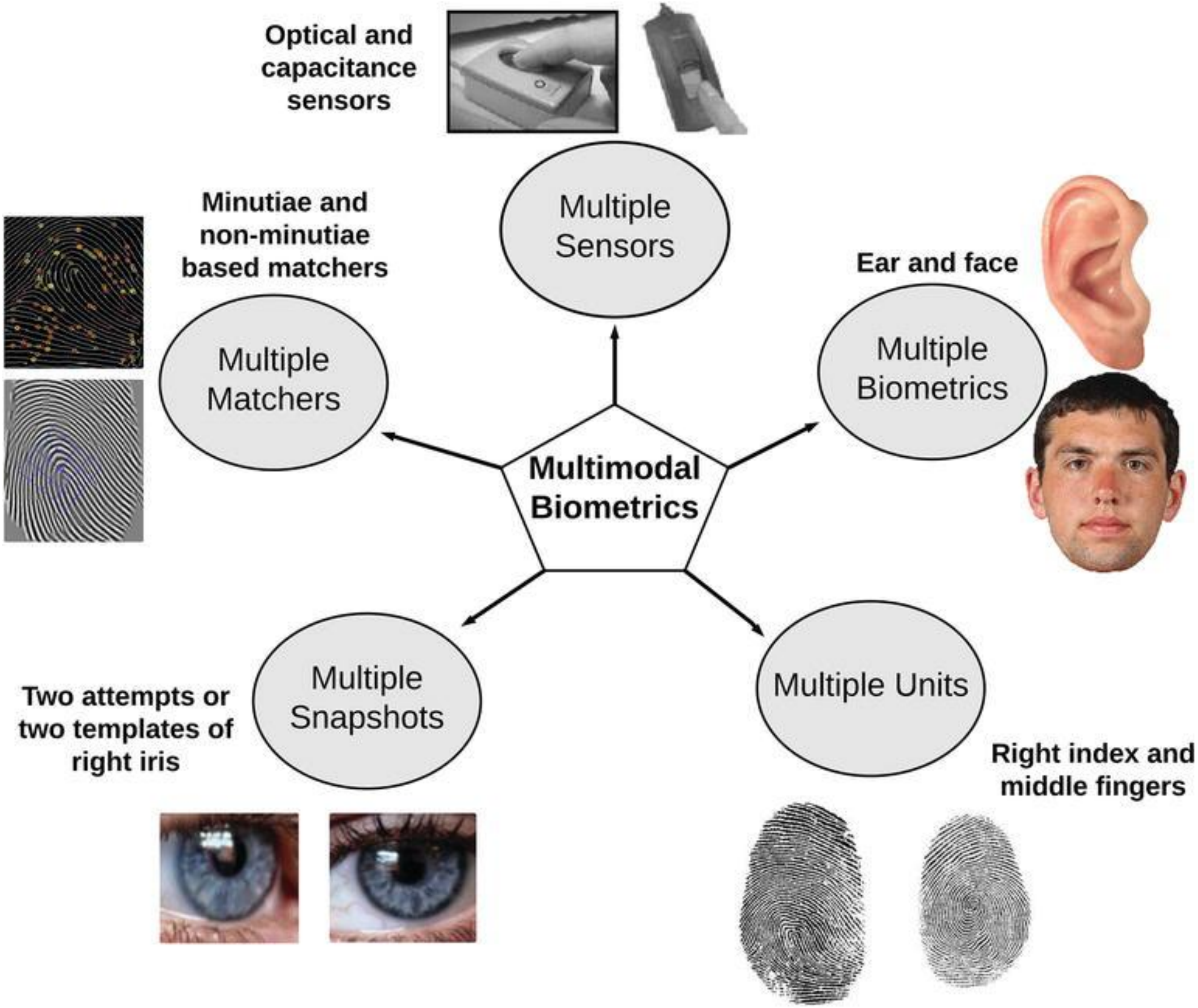
- **Such inputs can be expressed in different terms**

Taxonomy

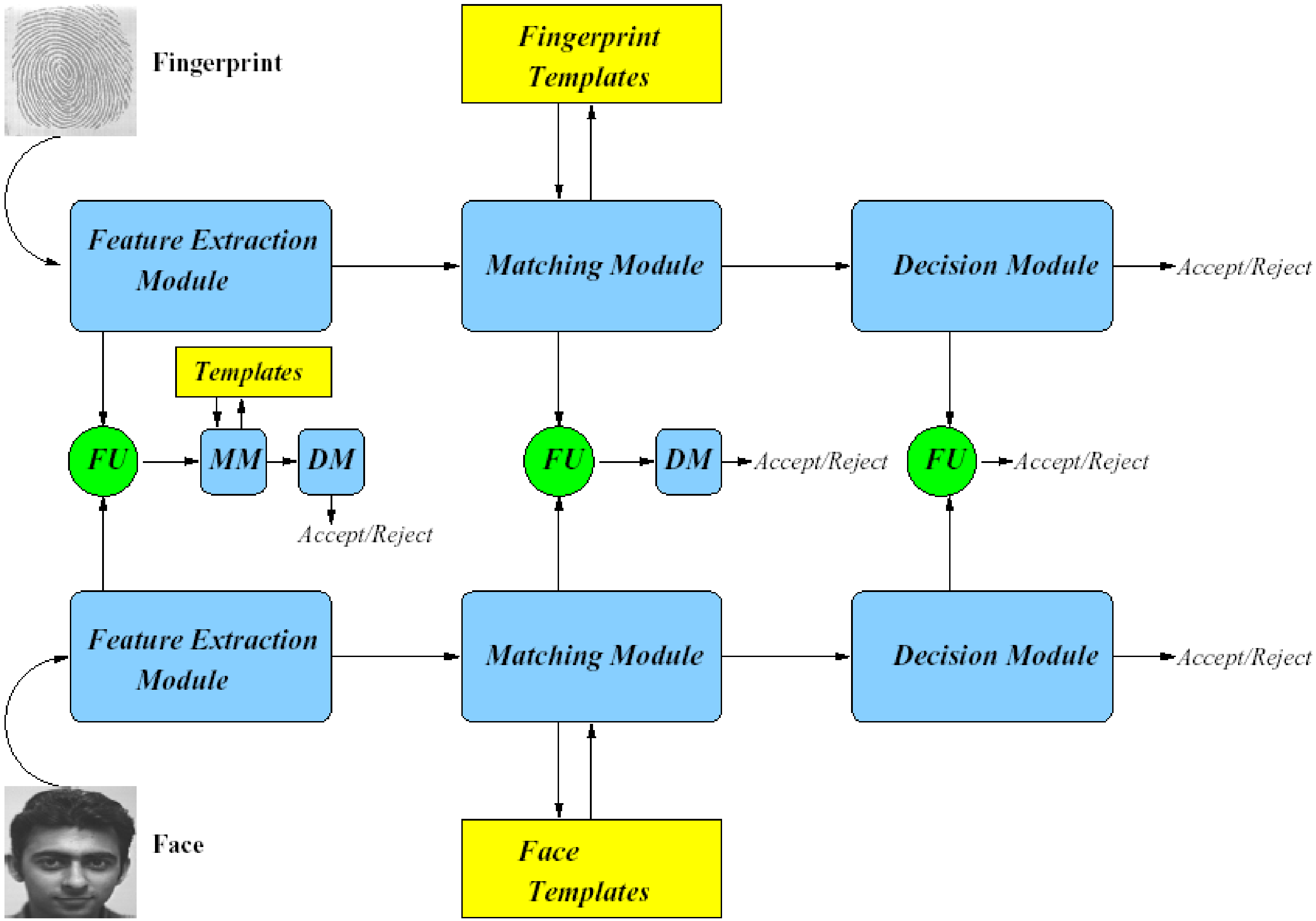
- **Combination can be referred to**
  - multiple instances of the samebiometric
  - multiple instances of multiple biometrics

- **It can be also referred to**
  - features adopted
  - computed matching scores
  - the estimated class

Multiple is better than single



Information-level fusion



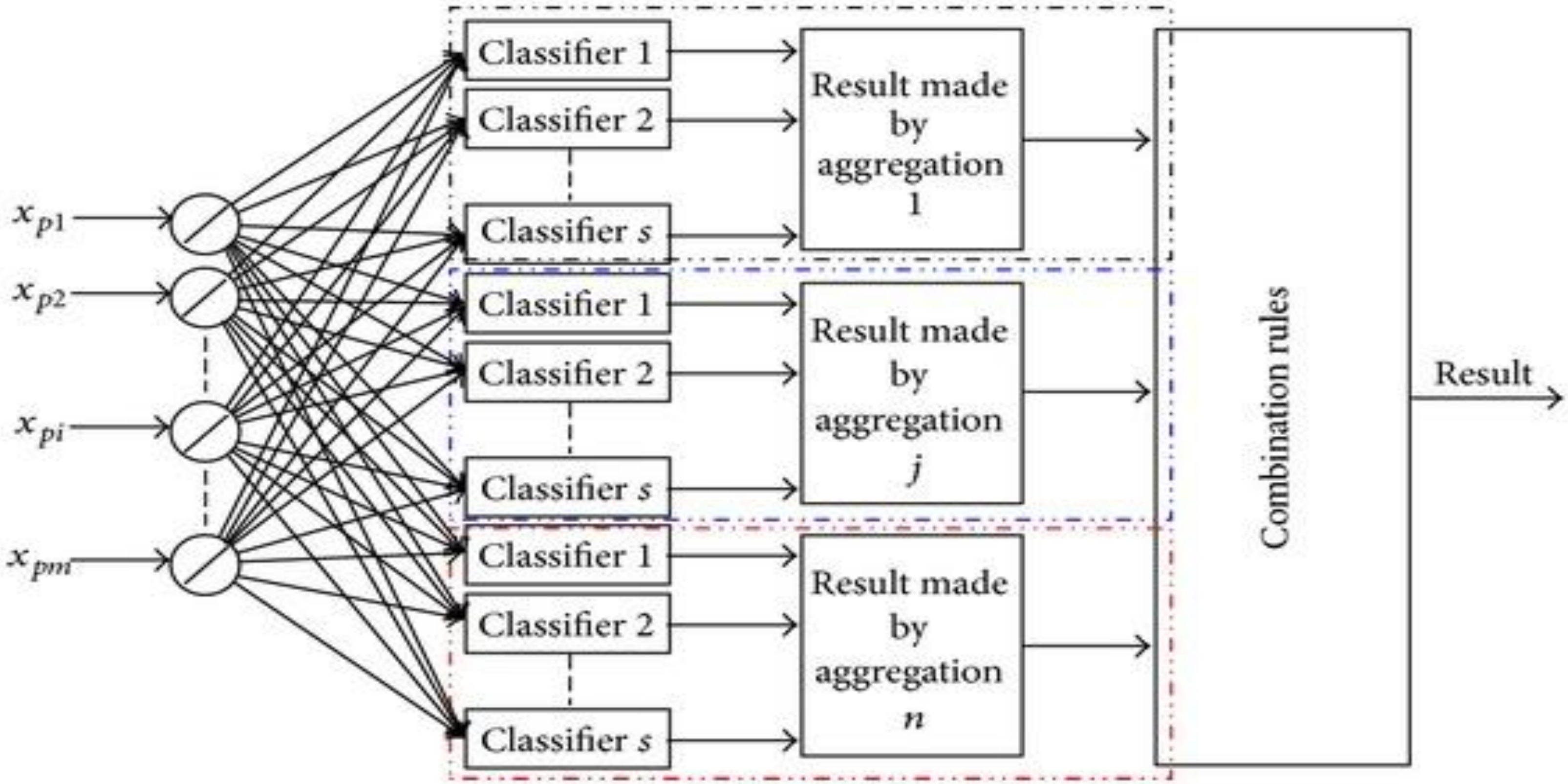
Rationale

- **The individual approaches are not able to meet the stringent requirements of many biometric applications**

- **E.g. face recognition systems:**
  - to handle the large variability of critical parameters, like
    - pose, lighting, scale;
    - face expression;
    - some kind of forgery in the subject appearance (e.g., beard)

- **On the other hand, multiple biometric recognition systems can be "complementary"**

Multimodality and multiple classifiers



Each level provide an «improved» evaluation of P(class|input) according to the a posteriori probability verification test

Potential of the fusion

- **Let us consider the following two biometric recognition systems performances based on the face:**
  - System 1 recognition rate: 85% (white and black population)
  - System 2 recognition rate: 80% (white and chinese population)
  - Faces correctly recognised by both Systems: 75% (white pop.)
  - Faces correctly recognised by System 1: 10% (black pop.)
    - And wrongly recognised by System 2
  - Faces correcly recognised by System 2 : 5% (chinese pop.)
    - And wrongly recognised by System 1
  - ***Potential recognition rate of both systems: 90%***
- **Therefore, the recognition rate can be significantly increased by an appropriate approach aimed to the *fusion* of the individual systems**

Fusion of multiple classifiers

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- **Measurement-level**
  - Posteriors of each class are combined by a more or less "combination functions"
- **Example**
  - Bi-modal classification (A/B) by feature-vector x
  - Two classifiers C1 and C2 output $p_{C1}(A|x)$ and $p_{C2}(A|x)$
  - Posteriors can be better estimated by *averaging* the above probabilities: $p(A|x)=0.5*(p_{C1}(A|x)+p_{C2}(A|x))$
  - Combination function: average among posterior probabilities
- **It is possible to proof that,**
  - in the case of simple and weighted average of posteriors,
  - and under the hypothesis of low correlation degree among estimations of posteriors,
  - such "fusion" increases the classification rate (accuracy)

Case-study 1: fingerprint classification

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$$\left\{p_A^{RNN}, \ldots, p_W^{RNN}\right\}$$

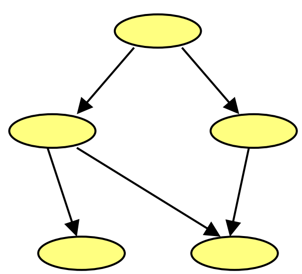
"Stacked" approach:
K-NN classifier for the final posteriors estimation

$$\left\{p_A^{MLP}, \ldots, p_W^{MLP}\right\}$$

RNN

MLP

Classifiers outcomes are estimates of posterior probability of the classes given the pattern



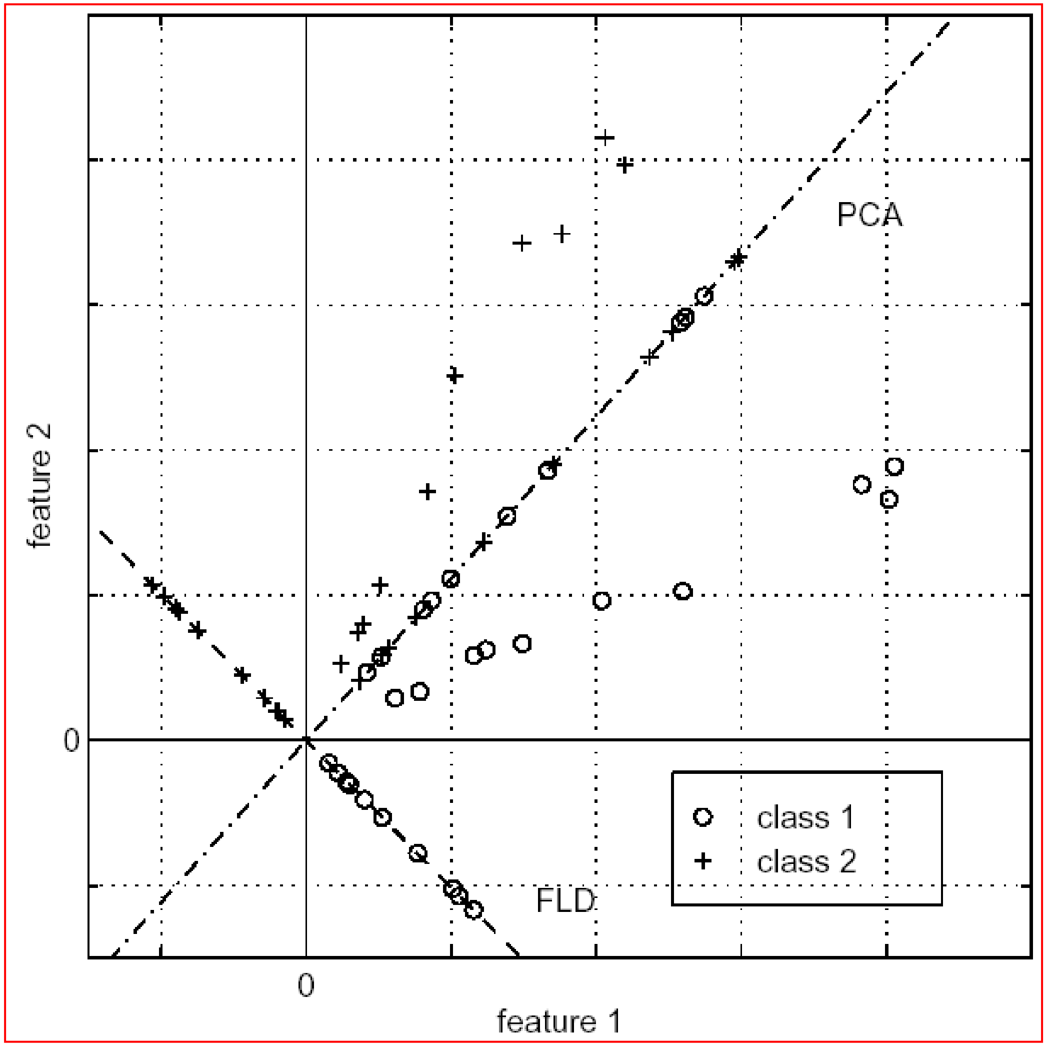
$(x_1, x_2, x_3, x_4, \ldots, x_N)$

For example, the "finger-code"

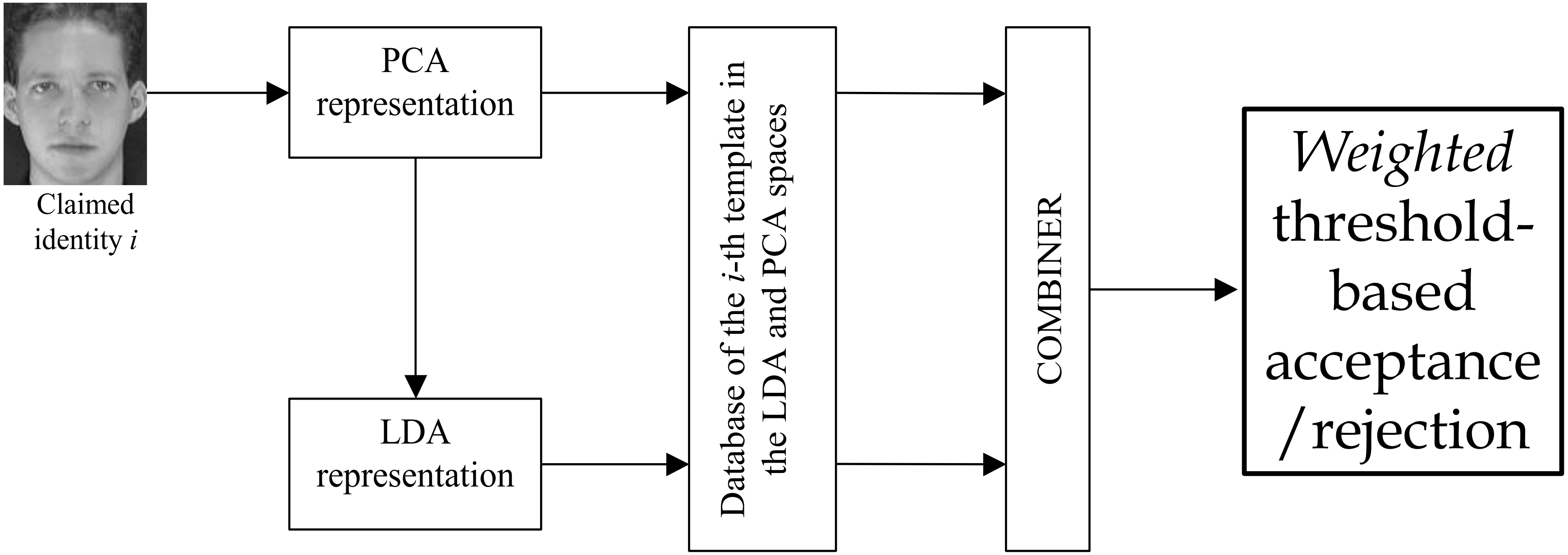
Multi-algorithmic fusion

- **Multiple matchers of the same biometric**
- **Fingerprint matching**
  - Fusion of texture and minutiae-based matchers
- **Face matching**
  - Fusion of multiple appearance-based methods
  - Fusion of appearance-based and structural matchers

Case-study 2: face recognition



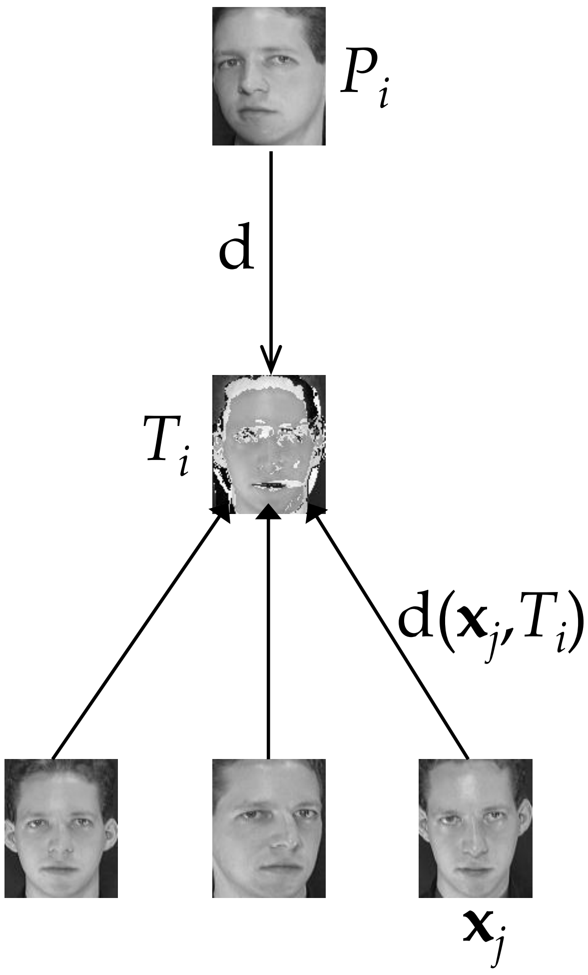
Exploiting PCA/LDA low correlation



Two-level fusion:

- Matching scores
- Threshold

Fusion approach



The identity i associated to the pattern P is accepted if:

$$\boxed{d(P, T_i) < w_i \cdot \lambda}$$

Threshold-level and distance-level fusion

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"Mean" algorithm: $w_i = \frac{w_i^{PCA} + w_i^{LDA}}{2}; \quad d = \frac{d^{PCA} + d^{LDA}}{2}$

"Max" algorithm: $w_i = \max\{w_i^{PCA}, w_i^{LDA}\}; \quad d = \frac{d^{PCA} + d^{LDA}}{2}$

"MaxMin" algorithm:

$$w_i = \max\{w_i^{PCA}, w_i^{LDA}\};$$

$$d = \min\{d^{PCA}, d^{LDA}\}$$

Decision-level fusion

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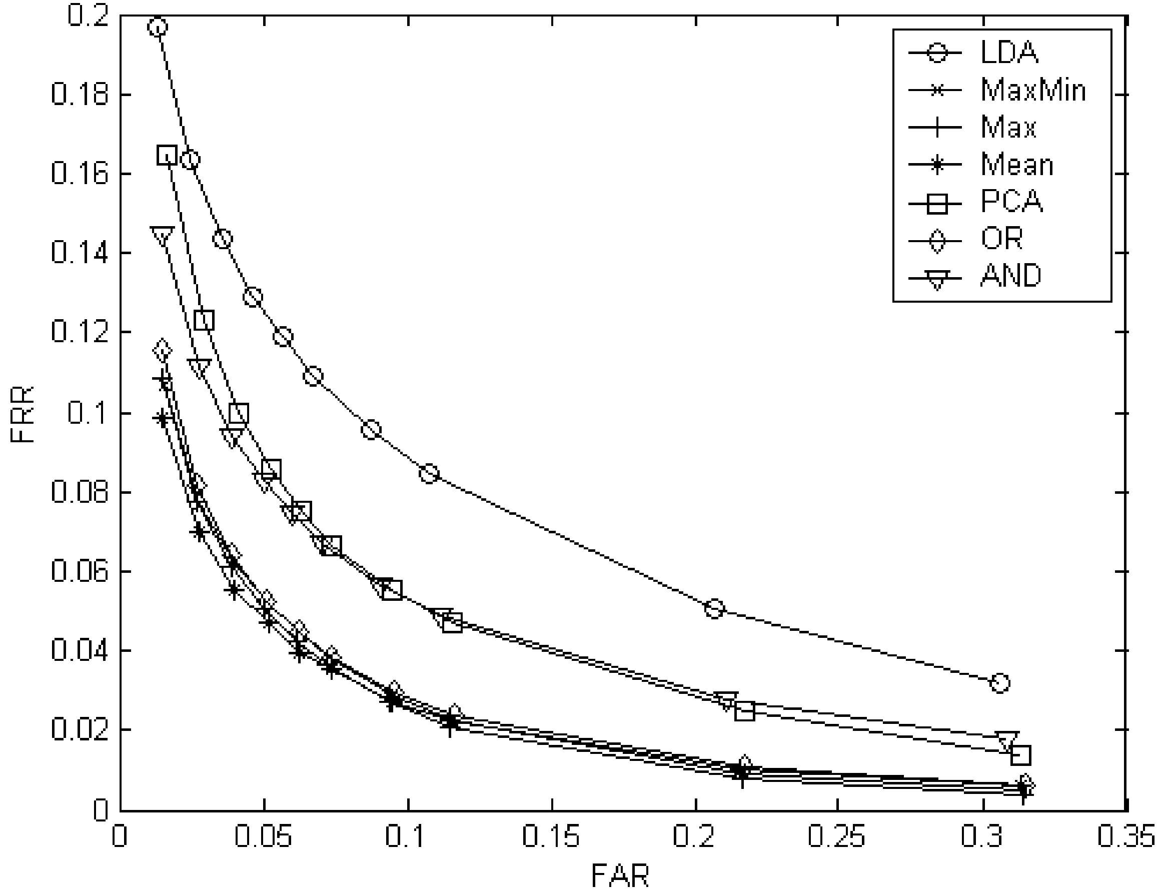
- **AND rule**
  - Accept = AND(Accept_PCA, Accept_LDA)
- **OR rule**
  - Accept = OR(Accept_PCA, Accept_LDA)
- **A lot of thresholds to be computed…**

ROC curves



LDA
MaxMin
Max
Mean
PCA
OR
AND
FRR
FAR
0.2
0.18
0.16
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35

Multi-sensor fusion

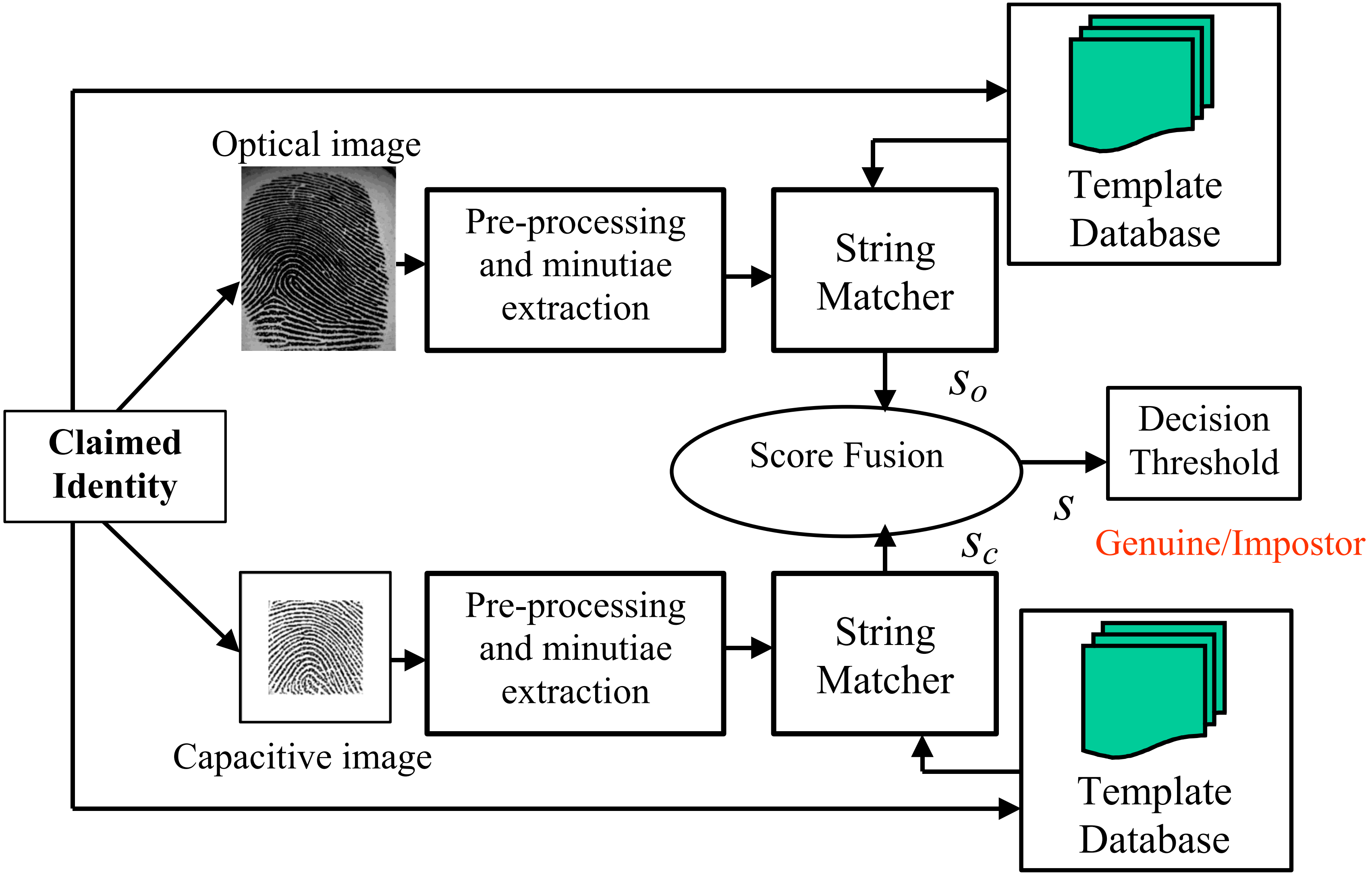
- **Using multiple cameras or multiple biometric scanners can lead to different images**
- **These images may allow to extract complementary features**
- **A case study in fingerprint verification: score-level fusion from images coming from optical and capacitive sensors**

Optical Sensor

Normal pressing, dry finger

Capacitive Sensor

A possible architecture



Measurement-level fusion rules

- **The couple of scores s_o and s_c provided by the optical and capacitive processing chains are fused using simple decision-level rules**

- **Example of "fixed" rule:**

$$s = \frac{s_o + s_c}{2}$$

- **Example of "trained" rule:**

$$s = \frac{1}{1 + \exp\left[-\left(w_0 + w_1 s_o + w_2 s_c\right)\right]}$$

Data set for experiments

Optical images

Capacitive images

Results

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	EER (s*)	FAR (s*)	FRR (s*)
Optical	3.4%	3.2%	3.6%
Capacitive	18.5%	18.2%	18.8%
Fusion by Mean	2.9%	3.1%	2.8%
Fusion by Logistic	2.3%	2.4%	2.3%

	Recover Rate Optical Sensor		Recover Rate Capacitive Sensor	
	Genuine	Impostor	Genuine	Impostor
Fusion by Mean	87.1%	91.7%	93.6%	86.7%
Fusion by Logistic	75.8%	70.7%	97.9%	94.3%

Recovering false rejections

Rotation

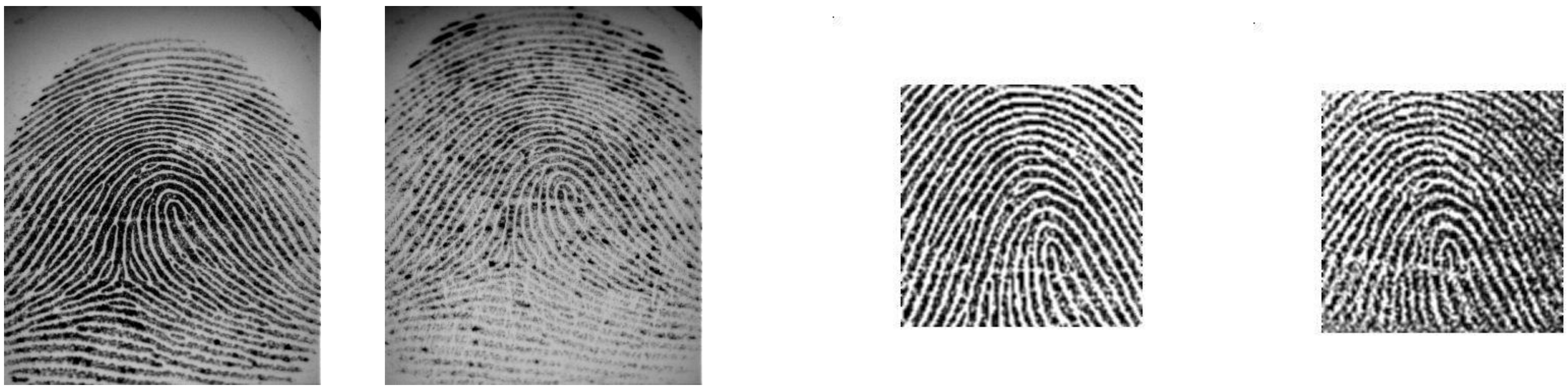
Rejected

Accepted

Accepted by fusion

Recovering false rejections

Bad pressing



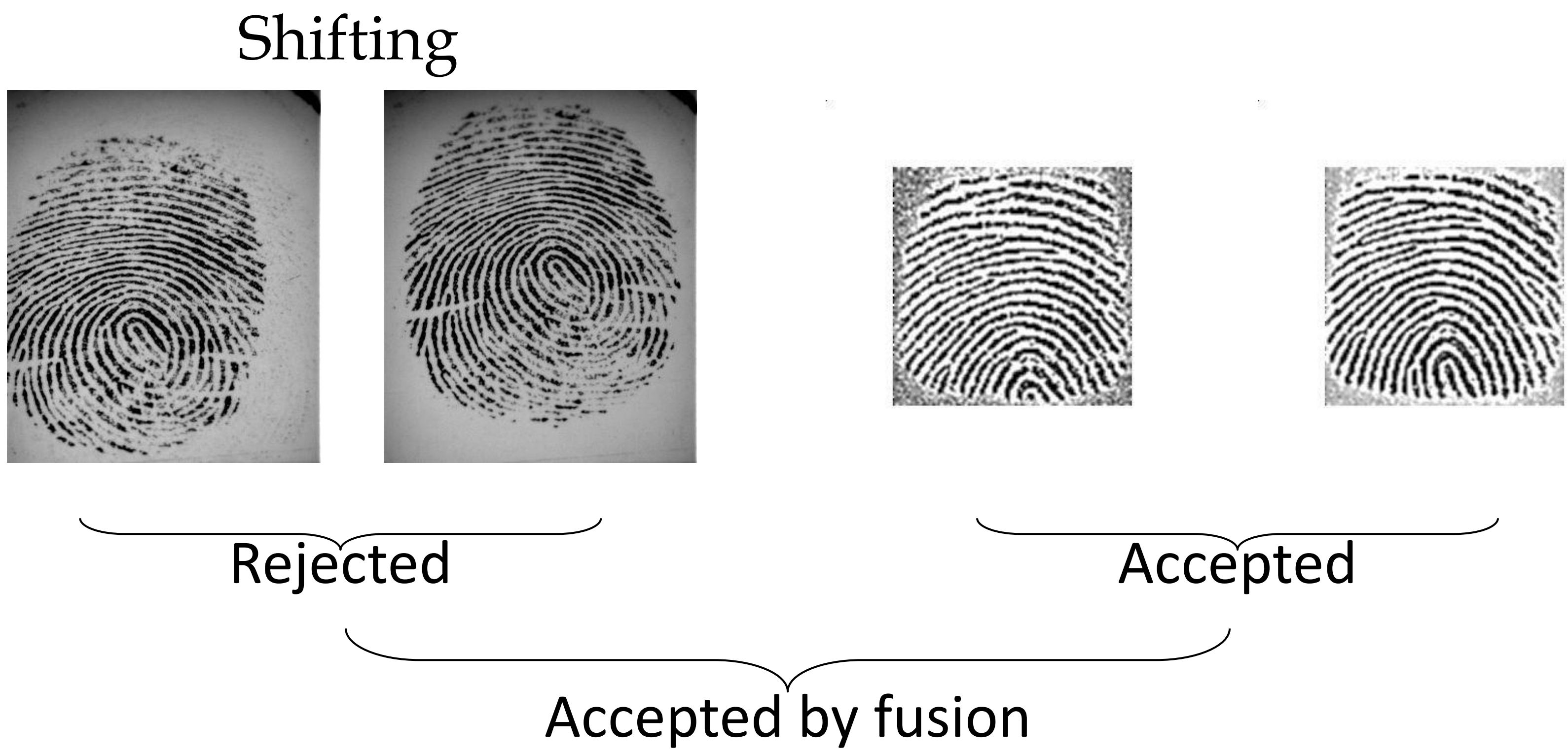
Rejected

Accepted

Accepted by fusion

G.L. Marcialis and F. Roli, Fingerprint verification by fusion of optical and capacitive sensors, *Pattern Recognition Letters* , Elsevier, 25 (11) 1315-1322, 2004.

Recovering false rejections



Multi-modal fusion

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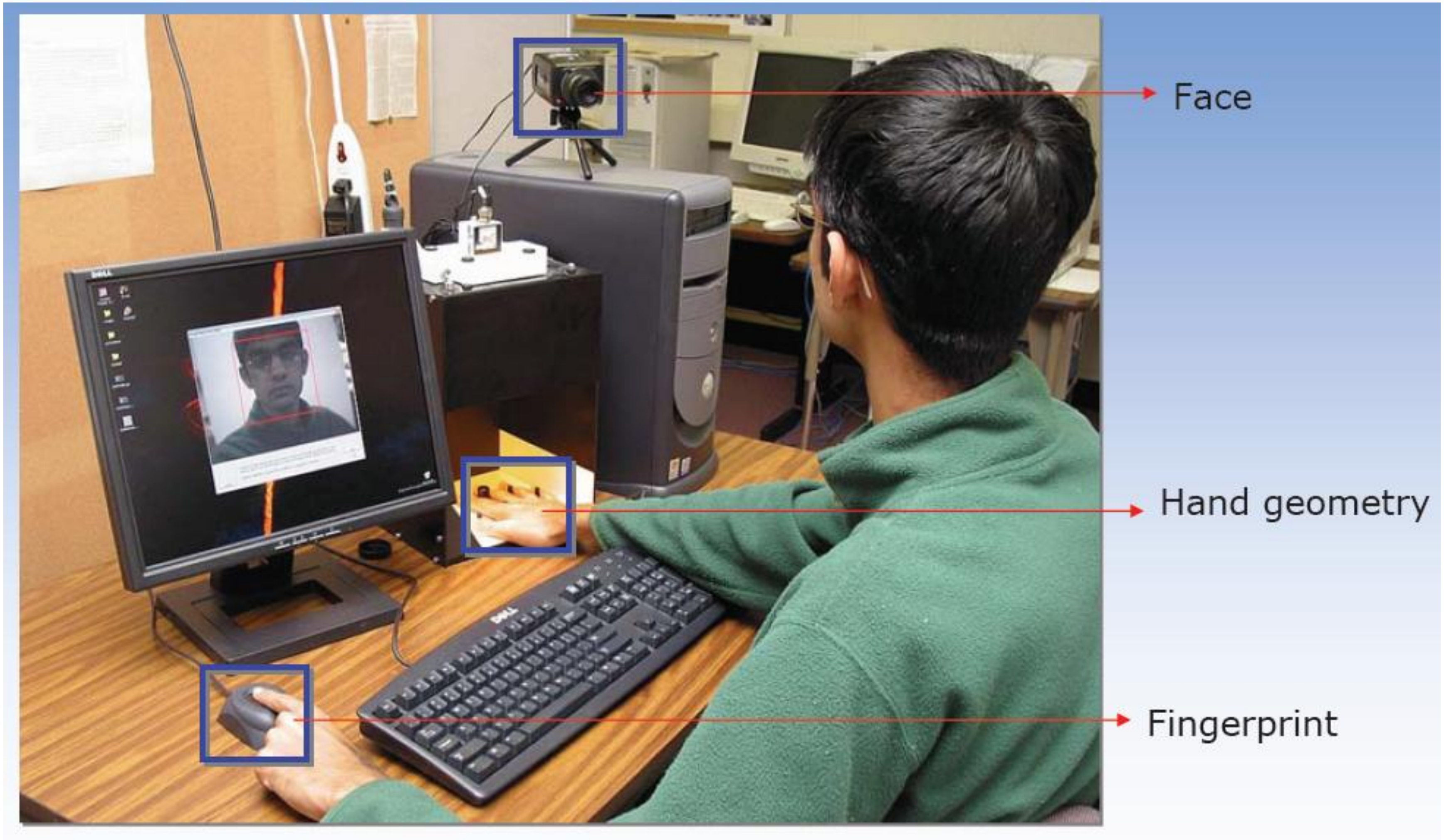

An example

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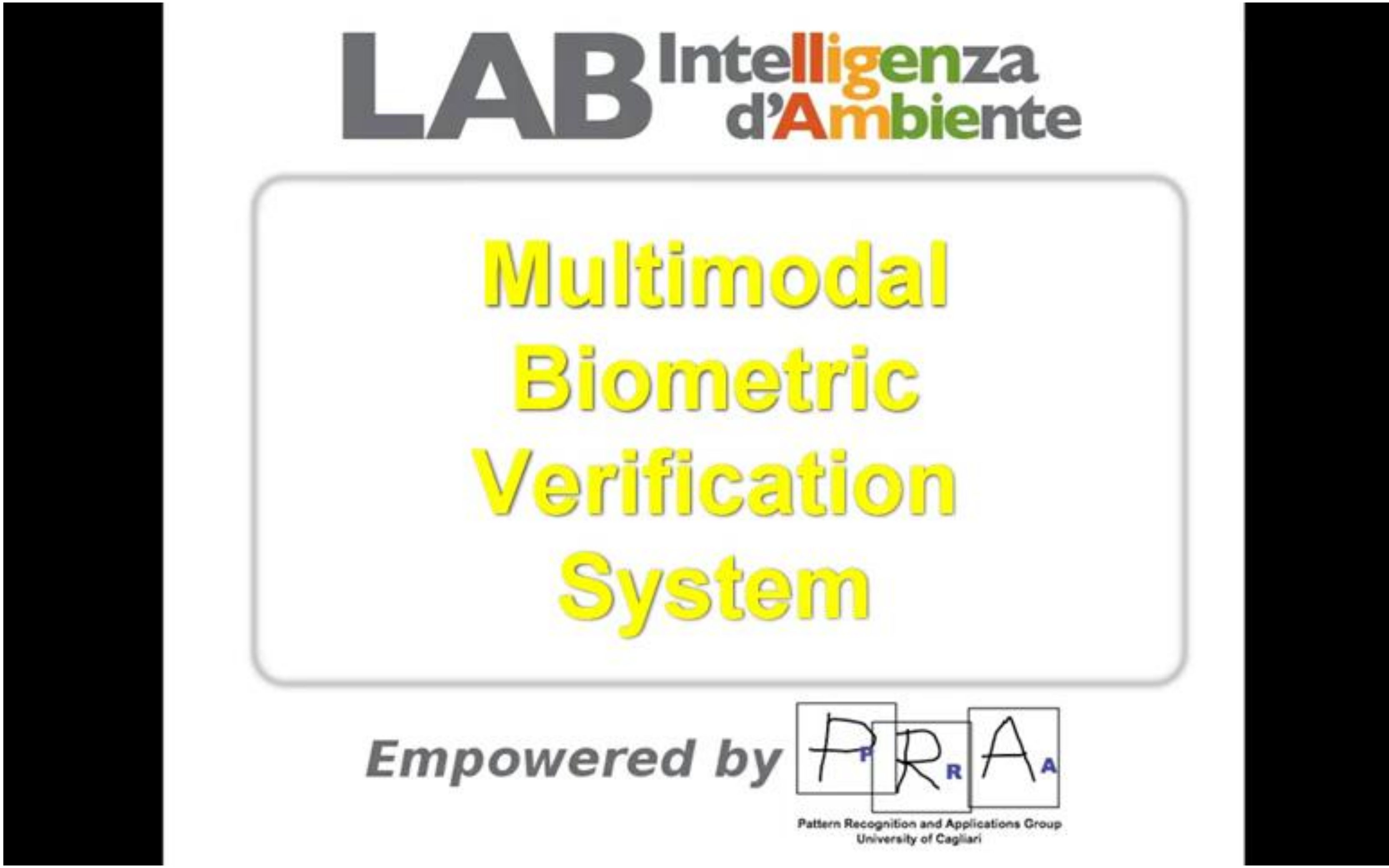

Performance, acceptability, cost

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	Universality	Uniqueness	Permanenence	Collectability	Performance	Accettability	Circumvention	Cost	Maturity
Fingerprint	M	H	H	M	H	M	L	M	H
Face	H	L	M	H	L	H	H	L	M
Iris	H	H	H	M	H	L	L	H	M
Hand vein	M	M	M	M	M	M	H	H	L
Facial thermogram	H	H	L	H	M	H	H	H	M
Retinal scan	H	H	M	L	H	L	H	H	H
Ear	M	M	H	M	M	H	M	M	L
Hand geometry	M	M	M	H	M	M	M	L	H
Voice	M	L	L	M	L	H	H	L	H
Signature	L	L	L	H	L	H	L	M	M
Gait	M	L	L	H	L	H	M	H	L

Serial fusion or cascade-based systems

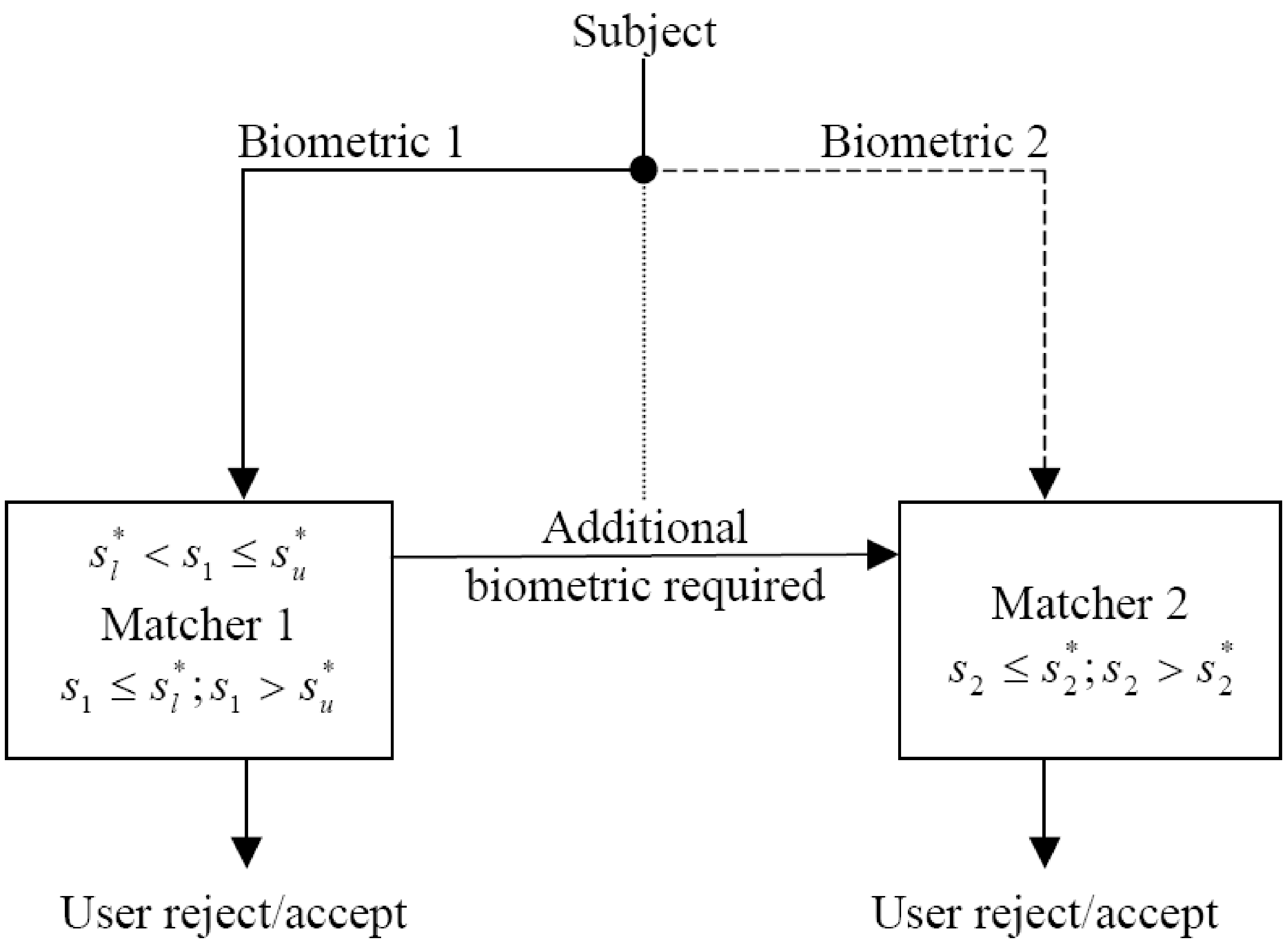
- **The previous approaches refer to the “parallel” fusion of biometric systems**

- **Drawbacks**
  - They increase the system invasiveness and the cooperation required
  - Some genuine users could be accepted by using only one biometric trait
  - The average verification time is always equal to that of the slowest biometric system

Approaches

- **Threshold-based**
  - The processing chain is set accordingly
- **Hypothesis testing-based**
  - Sequential Probabilty Ratio Test (SPRT), also called Wald's test

Threshold-based fusion



Theoretical modelling

- **Under the hypothesis that first matcher is working at zeroFRR-zeroFAR operational points, performance of the serial system is as follows:**

$$FAR = zeroFRR_1 \cdot FAR_2$$

$$FRR = zeroFAR_1 \cdot FRR_2$$

Serial system performs *better* than individual ones

In order to have performance gain, it is *sufficient* to collocate the worst matcher at the first stage

Results

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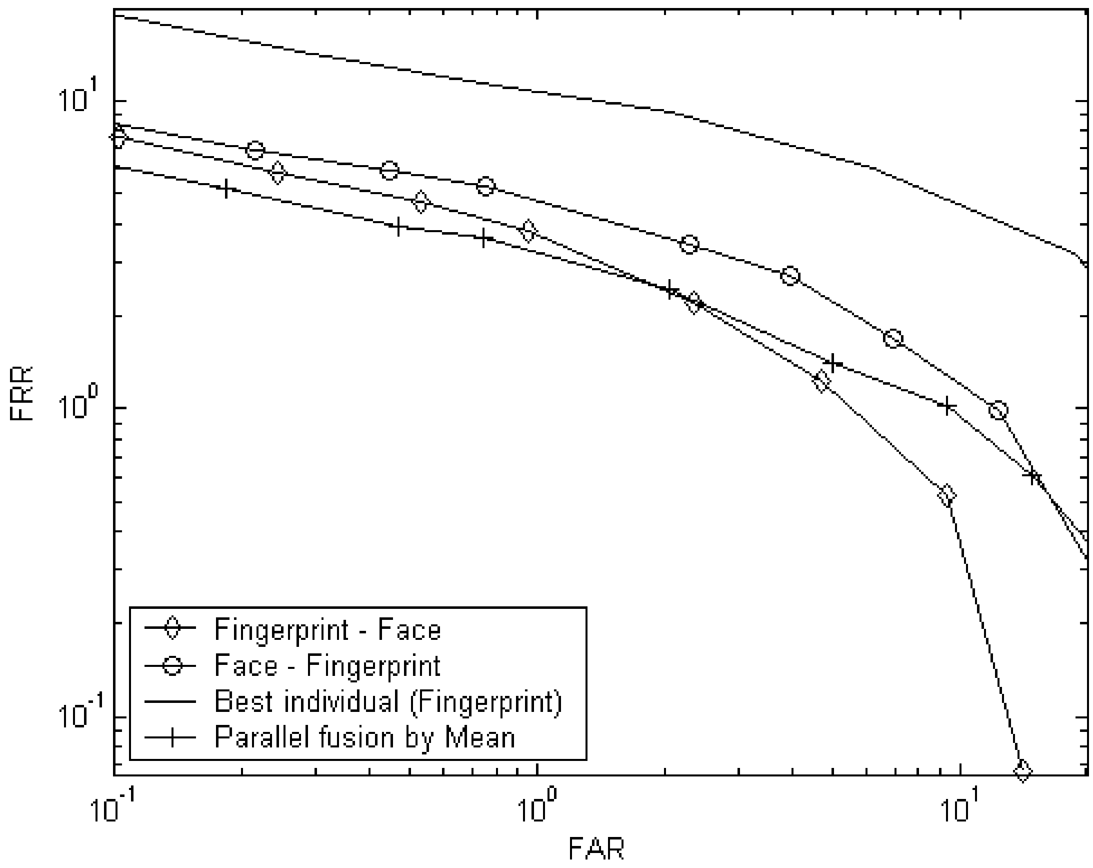
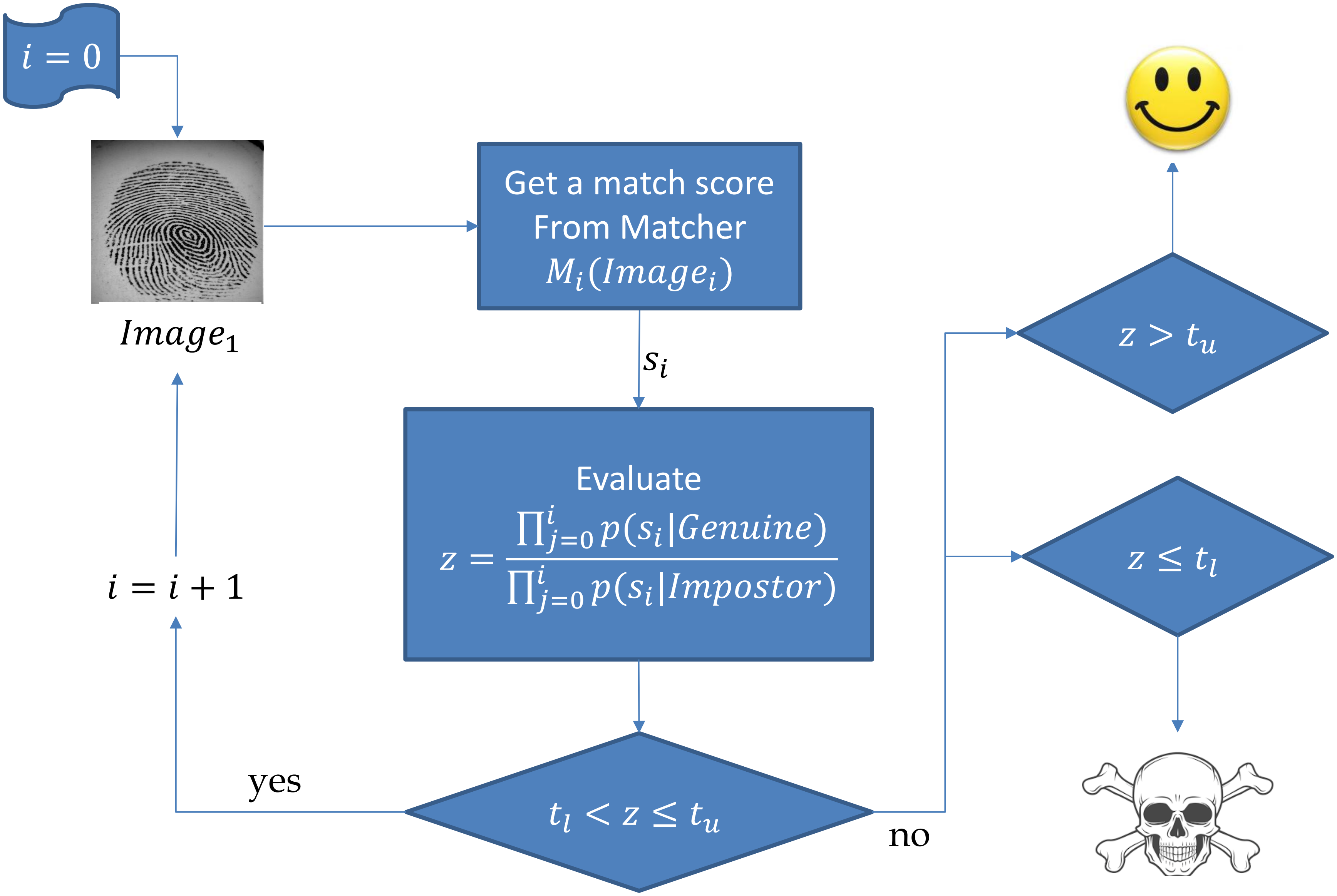


- M. Micheletto, G.L. Marcialis, G. Orrù, F. Roli, Fingerprint recognition with embedded presentation attacks detection: are we ready?, *IEEE Transactions on Image Forensics and Security*, https://arxiv.org/abs/2110.10567, DOI: 10.1109/TIFS.2021.3121201, 16 5338-5351, 2021.
- G.L. Marcialis, P. Mastinu, and F. Roli, Serial fusion of multi-modal biometric systems, Proc. of IEEE International Workshop on Biometric Measurements and Systems for Security and Medical Applications (BioMS2010), September, 9, 2010, Taranto (Italy), ISBN: 978-1-4244-6302-2, DOI: 10.1109/BIOMS.2010.5610438, pp. 1-7.
- G.L. Marcialis, F. Roli, and L. Didaci, Personal identity verification by serial fusion of fingerprint and face matchers, *Pattern Recognition*, Elsevier, doi:10.1016/j.patcog.2008.12.010, 42 (11) 2807-2817, 2009.

The Wald's test for biometric fusion



A toy example

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Tab.1. Accuracy of matching subsystems

Subsystem	A	B	C	D	E
FMR	30%	20%	10%	5%	1%
FNMR	30%	20%	10%	5%	1%

Tab.2. Total accuracy for different order of input

Input order	A→B→D→E→C	A→E→C→B→D
FAR	0.06%	0.38%
FRR	3.5%	1.7%

The suitability of SPRT (Sequential Probability Ratio Test) depends on the reliability of individual estimations of p(s|Genuine) and p(s|Impostor)

Multimodal fusion and liveness detection

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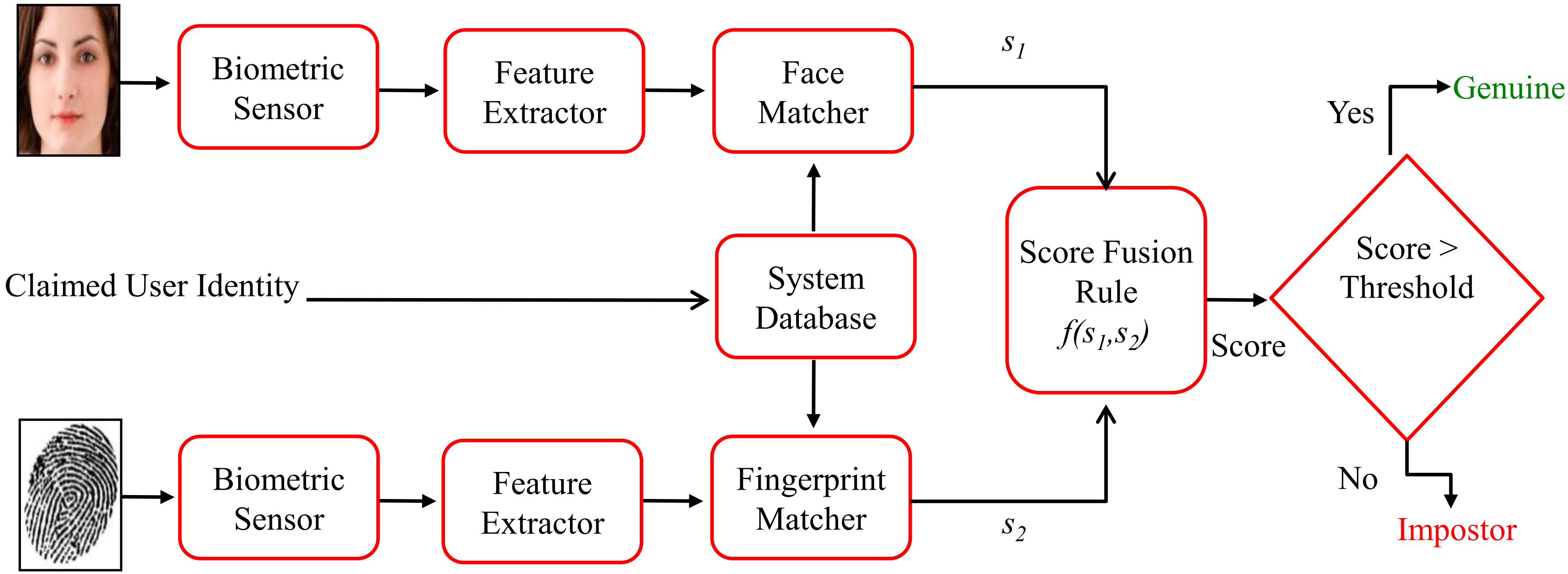
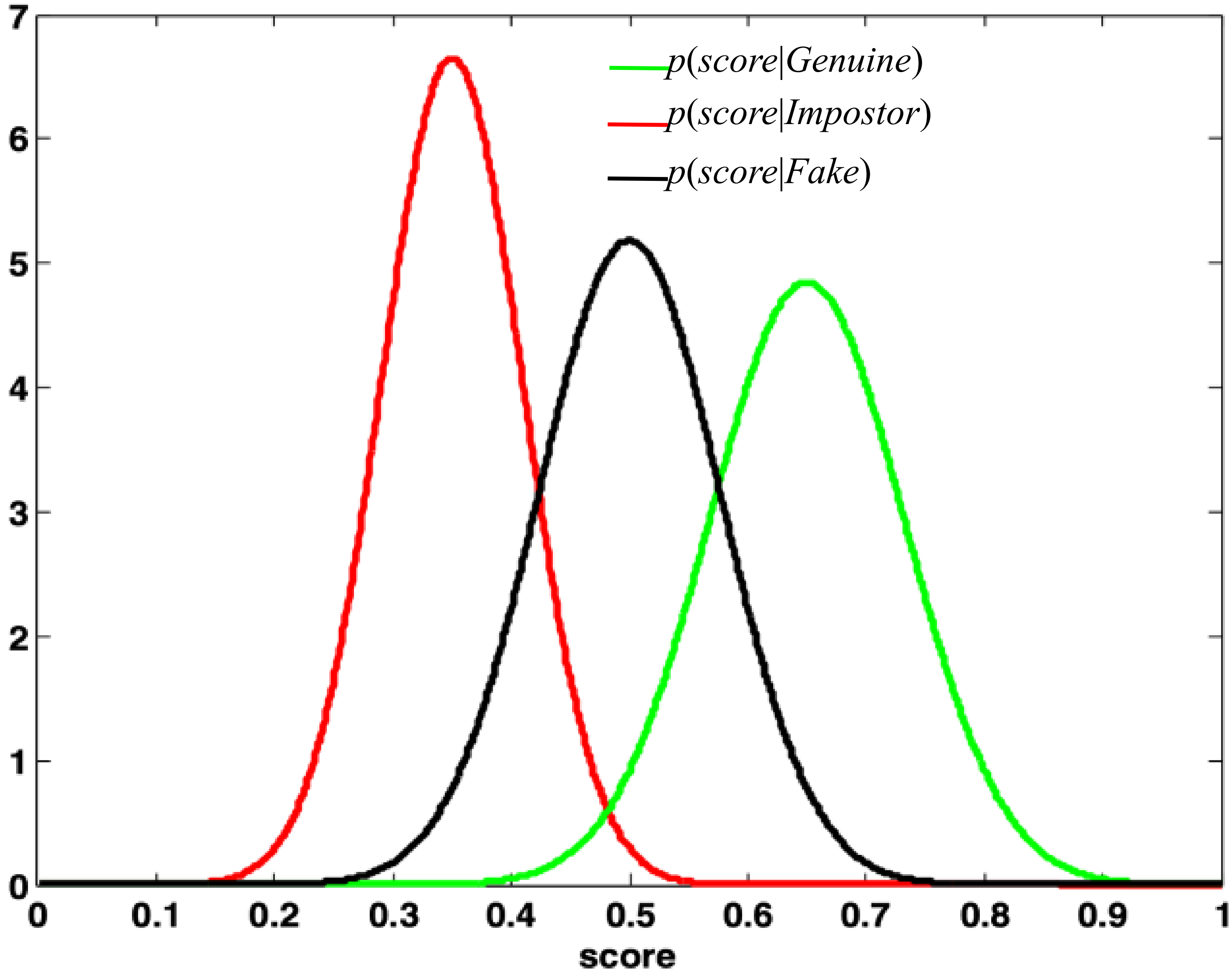


- **Is it necessary to spoof both biometrics?**
- **Can we simulate a possible multimodal spoofing attack?**

Relationship between live and fake distributions of match scores

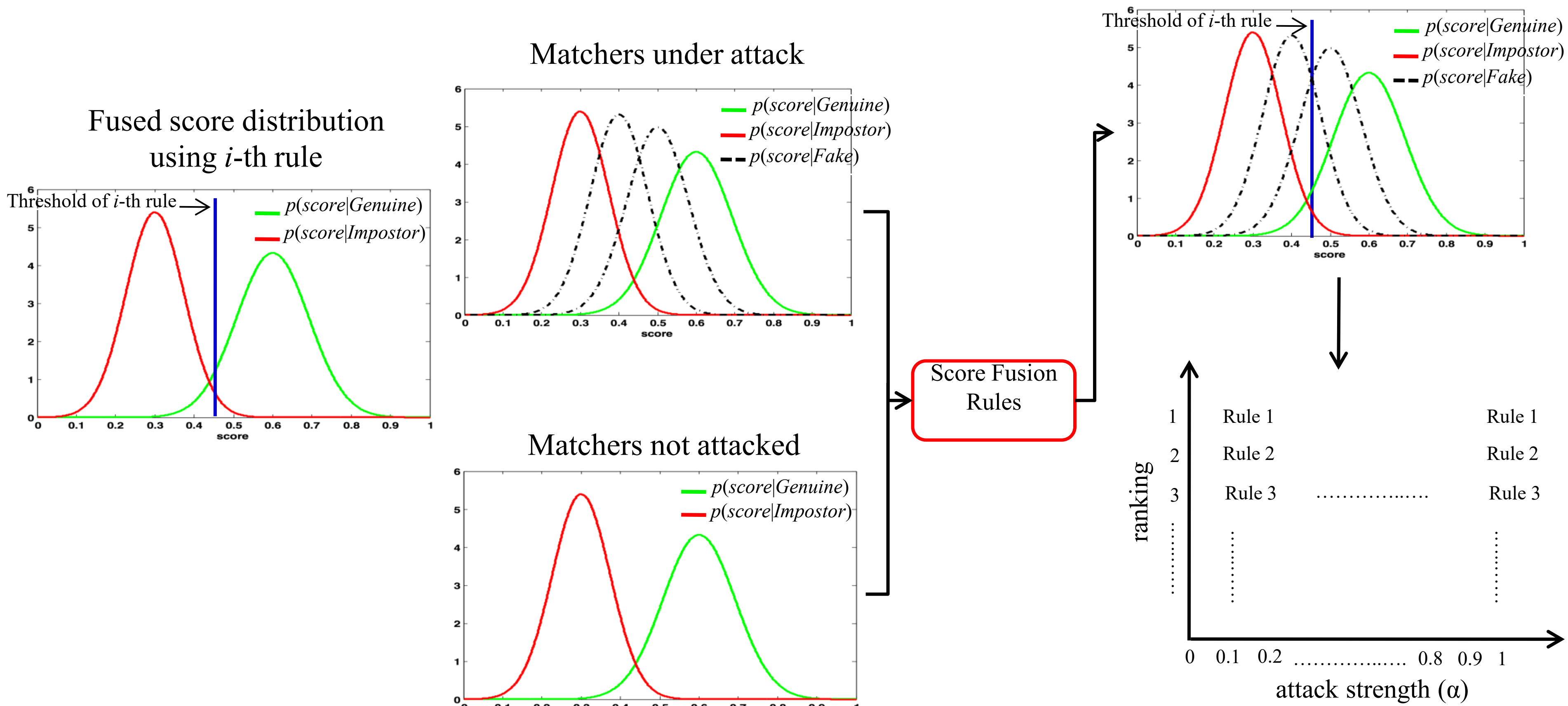


«Mimicking» spoofing distributions

Training Phase

Testing Phase

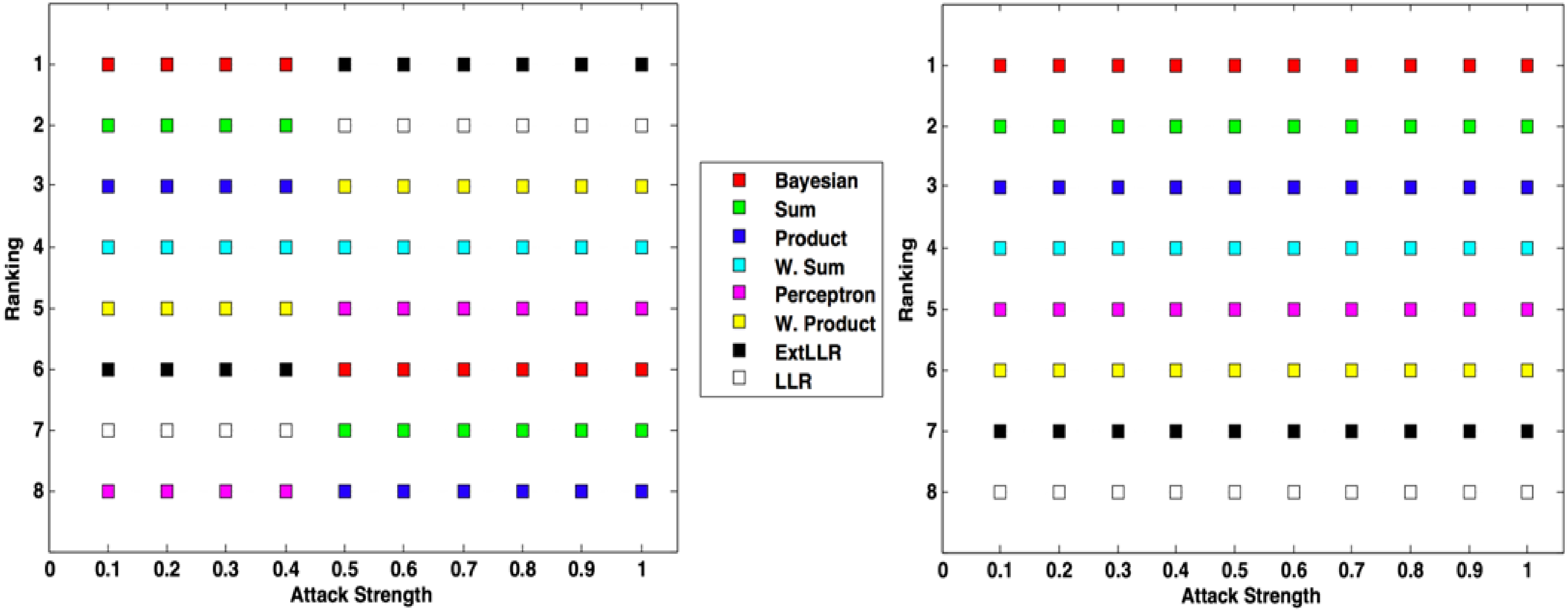
$$score_{Fake} = (1 - \alpha)\ score_{Impostor} + \alpha\ score_{Genuine}$$



Results

- **Fingerprint matcher is much more accurate than face matcher**

Face spoofing ($\alpha_{best} = 0.91$) Fingerprint spoofing ($\alpha_{best} = 0.52$)



$$score_{Fake} = (1 - \alpha)\ score_{Impostor} + \alpha\ score_{Genuine}$$

General purpose fusion rules

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- **Fixed : Sum, Product, Bayesian rule**

$$s = \frac{s_1 \cdot s_2}{s_1 \cdot s_2 + (1 - s_1) \cdot (1 - s_2)}$$

- **Trained : Weighted Sum, Weighted Product**

- **Stacked (by optimization)**
  - Perceptron/Logistic
  - Log-Likelihood (LLR)

- **«Spoofing-specific» rule: Extended LLR**

Specific fusion rules: extended LRR

$$\lambda = \frac{p(s_1, s_2|G)}{p(s_1, s_2|I)}$$

$$\begin{aligned} p(s_1, s_2|I) &= \frac{\alpha}{3}(1-c_1)(1+c_2)p(s_1|G)p(s_2|I) \\ &+ \frac{\alpha}{3}(1+c_1)(1-c_2)p(s_1|I)p(s_2|G) \\ &+ \frac{\alpha}{3}(1-c_1)(1-c_2)p(s_1|G)p(s_2|G) \\ &+ \frac{\alpha}{3}(c_1+c_2+c_1c_2)p(s_1|I)p(s_2|I) \\ &+ (1-\alpha)p(s_1|I)p(s_2|I)\,. \end{aligned}$$

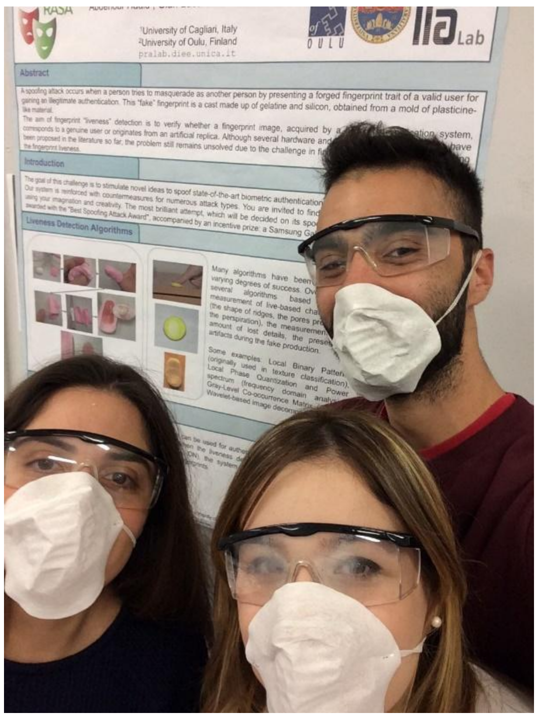
To take into account the probability of attacking individual matchers

That’s all for today… what’s next?

- **Soft Biometrics**
- **Behavioural Biometrics**
  - properties
  - pros and cons

- **Facial expression recognition**
- **Signal-based biometrics: EEG (ElectroEnceloGraphy)**

Thank you for listening!



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